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Drug Safety in Pregnant Women and Their Babies: Ignorance Not Bliss

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Although clinical trials address questions regarding drug safety for most segments of the population, pregnant women constitute one special group that is “orphaned” with respect to this issue. The lack of adequate pregnancy safety information for the vast majority of medications, combined with a need to make appropriate treatment decisions and to communicate risk information to a potentially vulnerable population, are some of the most challenging and critical women’s health issues.

The problem is widespread. One large US study estimated that 64% of women are being prescribed one or more drugs, not including vitamins and minerals, during pregnancy.¹ In part, this is due to inadvertent exposures to medications that occur before a woman knows that she is pregnant, as approximately half of all pregnancies are unplanned,² and <50% of pregnancies are recognized before the 4th week after conception.³ Intentional exposures throughout gestation are also common, and treatment of acute or chronic conditions such as hypertension, depression, or asthma may be necessary to protect the health of both the woman and her fetus or infant. Thus, the clinician and the pregnant patient are frequently faced with the dilemma of assessing the potential risk of exposure to a medication that occurred early in a pregnancy or of making the decision to initiate, continue, or modify a medication regimen during pregnancy or breastfeeding.

Unfortunately, however, sound data on which to base these decisions are frequently lacking. The preclinical reproductive toxicity studies in animal models that are required for regulatory approval of most drugs are not completely predictive of human risk, and randomized clinical trials of medications usually cannot be ethically justified. After a medication has been approved for clinical use, manufacturers are usually only required to collect adverse pregnancy outcomes that become known through voluntary case reports, a system that is very incomplete and difficult to interpret. Formal controlled observational studies of outcomes of exposed pregnancies are conducted in a piecemeal

fashion for only a small fraction of available medications through the uncoordinated efforts of various individual investigators or state or federal agencies. As a result, there are insufficient human pregnancy safety data available for >80% of 468 new drugs marketed over a recent 20-year period in the United States.⁴

As no systematic method for routinely performing these kinds of reproductive outcome studies currently exists, there is often a long delay in gaining adequate information about new drugs and their use during pregnancy. Furthermore, as pregnancy data begin to emerge, the interpretation of results is hampered by statistical difficulties inherent in studying rare outcomes such as birth defects and the challenges of limited sample sizes.⁵

In the last few years, these issues have played out in a particularly high profile manner, with a number of studies addressing the risks of selective serotonin reuptake inhibitors during pregnancy. Although several small human studies published in the 1990s suggested that first-trimester exposure to medications of this type does not greatly increase the risk for birth defects overall,^{6,7} sample sizes in these studies were insufficient to rule out increased risks for even the more commonly occurring specific birth defects, such as heart defects, neural tube defects, or oral clefts. More recently, some studies with larger sample sizes and more efficient study designs have suggested small but significantly increased risks for a few major structural defects and other rare outcomes.^{8–11} However, as no systematic and comprehensive method for evaluating new drugs exists in the United States, it has taken almost 20 years to obtain this level of information since the first drug in this class was marketed, despite the fact that selective serotonin reuptake inhibitors are among the most widely used prescription drugs ever developed.

The real-world situation is that the clinician and patient must make decisions with the incomplete data available at the moment; oftentimes in situations where no equally effective alternative medication or other treatment is available. Even if alternative treatment choices can be considered, they usually lack adequate information on safety during pregnancy as well. Nevertheless, these decisions must take into consideration the

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possible negative consequences to the mother and developing embryo or fetus if the mother discontinues medication, switches to a less effective medication, or is undertreated.^{12,13} Evaluating all of these factors to assess the risk for an individual patient is a complex task with many unknowns.

Evidence-based risk assessment also requires an understanding of the limits of the available data. Clinicians may not have the training or expertise needed to evaluate reproductive and developmental toxicity data critically and translate it to a clinical setting.¹⁴ Similarly, clinicians may not have sufficient training in statistics to place the appropriate confidence bounds around results from small studies reported in the literature. Instead, clinicians and the public most often rely on summary pregnancy safety information provided in the product label, such as the Food and Drug Administration's A, B, C, D, and X letter categories.

Although appealing in their simplicity, these pregnancy categories are almost exclusively derived from animal data and have been sharply criticized for failure to differentiate between severity of adverse outcomes, the range of adverse pregnancy outcomes, or the importance of dose, route, gestational timing, or indication.¹⁵ For example, both doxycycline and valproic acid are classified as pregnancy category D (indicating that "there is positive evidence of human fetal risk based on adverse reaction data from investigational or marketing experience or studies in humans, but potential benefits may warrant use of the drug in pregnant women despite potential risks"). However, doxycycline treatment is associated with a risk of staining the baby's teeth, and valproic acid treatment is associated with a high incidence of several specific major birth defects such as spina bifida. Clearly, these risks are quite different in severity despite classification in the same category D designation.

In the last decade, the Food and Drug Administration has developed and proposed rules for restructuring the "pregnancy label" (Table 1).¹⁶ This label revision responds to many of the concerns raised regarding the letter category designations by providing a structured narrative with more precise risk

descriptions as well as a summary of the underlying data (or lack thereof). Its implementation would be an important step in the right direction.

No less challenging for the clinician is communication of a risk assessment to a woman who is or who may become pregnant in a comprehensible manner that allows her to consider the issues appropriately.¹⁷ Although perceived small risks for uncertain outcomes may not alter behavior under usual circumstances, any perceived risk to the baby, no matter how small, may be unacceptable to women and their partners or physicians.¹⁸ A woman may view the benefit for herself as insufficient relative to the perceived risk or even the undefined possibility of risk to her infant. In some cases, wanted pregnancies may be terminated because of misperceptions regarding the presence or magnitude of risk to the developing baby.¹⁹

On the other hand, communication of risks associated with high-level teratogens, such as isotretinoin, also presents challenges bringing home the message in a manner that leads to the desired behavior, i.e., appropriate use of contraception or abstinence to avoid an exposed pregnancy. Specific information regarding the risks, coupled with specific information about the actions a woman can take to prevent pregnancy, must be delivered through a medium and in a manner that is both understood by the woman, and likely to lead to her compliance. Successful methods for risk communication and translation to behavior change may vary by age, cultural or educational background.²⁰ Clearly, more research is needed to determine best methods for communication in both high- and low-risk and high- and low-uncertainty situations.

However, by far, the overriding challenge in this area of women's health is obtaining adequate information on drug safety during pregnancy as quickly as possible after a new drug is marketed. Pregnancy registries are one method of collecting information on newly marketed medications. These registries are increasing in number, and more rigorous standards are being employed for their design.^{21,22} However, registries are not required for all new drugs. Furthermore, this approach alone cannot provide sufficient information to assess the risks for all major adverse pregnancy outcomes with an appropriate level of confidence. A national mandatory and systematic surveillance system for all newly marketed drugs used by women of childbearing age is imperative to deal with this issue. People of all ages are entitled to drug safety information for products marketed for their use. Women of childbearing age and their babies are no exception.

CONFLICT OF INTEREST

The authors declared no conflict of interest.

Table 1 Rules for restructuring the pregnancy label

Proposed rule changes to the pregnancy and lactation portions of the product label

1. Information on pregnancy registries, if any, for the product and how the registries can be contacted.
2. A general statement about background risk of adverse pregnancy outcome.
3. A "Fetal Risk Summary" containing a narrative description of the risks of use of the medication, including a characterization of the likelihood that the drug increases the risk of abnormal development including structural malformations, miscarriage, stillbirth, neonatal death, functional abnormalities, and alterations in growth. Conclusions about risk will use standardized statements and will differentiate between animal and human data. If a drug is contraindicated, the specific circumstances, such as timing, under which its use is contraindicated will be described.
4. A section on clinical considerations to address inadvertent vs. intentional exposures, risks of untreated disease, gestational timing, and dose considerations in pregnancy.
5. A summary of the data underlying the fetal risk and clinical consideration statements.

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